

JAMES R. VALCOURT

valcourt@g.harvard.edu ~ (978) 870-5393

SUMMARY

Interdisciplinary researcher with experience marshalling both experimental and computational techniques to tackle research problems with applications in drug discovery, development, and regenerative medicine.

EDUCATION

Harvard University: Cambridge, Massachusetts

- Ph.D. in Systems, Synthetic, and Quantitative Biology, 2014 – 2020 (expected)

Princeton University: Princeton, New Jersey

- A.B. in Molecular Biology, *magna cum laude*, 2012
- GPA: Cumulative, 3.89; Departmental, 3.91
- Certificate in Quantitative and Computational Biology

WORK EXPERIENCE

D. E. Shaw Research: New York, New York, USA

- Scientific Associate, 2012 - 2014
- Conducted long-timescale molecular dynamics simulations of pharmaceutically relevant biological systems
- Special focus on the mechanisms of allosteric modulation in GPCRs and on the structural basis of neutralization breadth of HIV antibodies

SELECTED RESEARCH EXPERIENCE

Predicting and controlling mesendoderm competence in hESCs, 2015-19

- Graduate student with Sharad Ramanathan at Harvard University
- Discovered and manipulated novel modes of cell state control
- Designed, executed, and communicated all phases of project

PUBLICATIONS (ACADEMIC)

Valcourt, J.R., Huang, R.E., Kundu, S., Venkatasubramanian, D., Kingston, R.E., Ramanathan, S. Changing the Waddington landscape to control mesendoderm competence. *bioRxiv* 867010 [Preprint]. December 6, 2019. <https://doi.org/10.1101/867010>

Bokoch, M. P., Jo, H. I., **Valcourt, J. R.**, Srinivasan, Y., Pan, A. C., Capponi, S., Grabe, M., Dror, R. O., Shaw, D. E., DeGrado, W. F., Coughlin, S. R. (2018). Entry from the lipid bilayer: a possible pathway for inhibition of a peptide G protein-coupled receptor by a lipophilic small molecule. *Biochemistry*, 57(39), 5748–58.

Dror, R. O., Green, H. F., Valant, C., Borhani, D. W., **Valcourt, J. R.**, Pan, A. C., ... Shaw, D. E. (2013). Structural basis for modulation of a G-protein-coupled receptor by allosteric drugs. *Nature*, 503(7475), 295–299.

Valcourt, J. R., Lemons, J. M. S., Haley, E. M., Kojima, M., Demuren, O. O., & Collier, H. A. (2012). Staying alive: Metabolic adaptations to quiescence. *Cell Cycle*, 11(9), 1680–1696.

PUBLICATIONS (TRADE)	Valcourt, J. R. “Systematic: How Systems Biology is Transforming Modern Medicine.” New York: Bloomsbury, 2017.
AWARDS AND DISTINCTIONS	– Peralta Award, Department of Molecular and Cellular Biology, Harvard University, 2019 – Quantitative Biology Fellowship, NSF-Simons Center for the Mathematical & Statistical Analysis of Biology at Harvard University, 2019 – National Defense Science and Engineering Graduate Fellowship, 2014 (declined) – National Science Foundation Graduate Research Fellowship, 2014 – Fannie and John Hertz Foundation Graduate Fellowship, 2012 – Moses Taylor Pyne Honor Prize, the highest general distinction conferred on an undergraduate by Princeton University, 2012 – Phi Beta Kappa, 2012 – Society of Sigma Xi, 2012 – Alberto Santos-Dumont Prize for Innovation (accepted on behalf of The Princeton Tiger Magazine), Princeton University, 2011 – Shapiro Prize for Academic Excellence, Princeton University, 2009
TECHNICAL SKILLS	Wet lab: mammalian cell culture, quantitative live cell microscopy, immunofluorescence, RNA/ATAC/ChIP-seq, fluorescence-activated cell sorting, microfluidics. Dry lab: molecular dynamics; numerical modeling; analysis of RNA/ATAC/ChIP-seq and genomic data, both bulk and single-cell; image analysis; statistical methods. Languages and Platforms: Python, Matlab, R, Linux, Slurm, LaTeX, some Tensorflow, some Julia, some Swift, Adobe Illustrator
TEACHING EXPERIENCE	Teaching Fellow, LS50 Integrated Science, Harvard University, 2016 Served as one of the inaugural teaching fellows for a new, intense double course for first-year undergraduates. Taught weekly sections, gave input on overall course structure, co-developed weekly problem sets, and co-wrote exams. Volunteer Tutor, Star Learning Center, New York City, 2013-2014 Taught high school chemistry for underprivileged students. Peer Tutor, Wilson College, Princeton University, 2009-2012 Tutored other students in entry-level science courses.
ADDITIONAL RESEARCH EXPERIENCE	Interactions between microRNA and RNA-binding proteins, 2010-2012 Worked with Professor Hilary Collier at Princeton University to investigate novel post-transcriptional regulatory networks

ADDITIONAL
RESEARCH
EXPERIENCE
(CONTINUED)

Quantifying *D. melanogaster* embryo morphogenesis, 2008-2010

Worked with Professor Eric Wieschaus at Princeton University to study the genetic and morphological processes of cephalic furrow formation and develop software to analyze morphogenesis in three dimensions

Hotspots of structural variation in the human genome, 2009

Worked with Dr. Jan Korbelt at the European Molecular Biology Laboratories to investigate the origins of differing structural variation rates in the genome

OTHER
ACTIVITIES

Harvard University: co-founder and co-organizer of the Quantitative Biology Series; Hertz Foundation in-school representative for Harvard and organizer of events and retreats for in-school fellows; miscellaneous public outreach including publishing a non-fiction book about systems biology intended for general audiences.

Princeton University: Chair of second oldest college humor magazine in the country; outdoor orientation program leader trainer and member of governing body; tour guide; Speakers Chair of Student Bioethics Forum.

Outdoors: Eagle Scout and Order of the Arrow, avid backpacker